

Omar Kahol

Italian (born 1998) • Budoia, PN • omar.kahol@yahoo.com • Website • Linkedin • GitHub

Engineer specializing in the intersection of Aerodynamics, Machine Learning, and UQ. Expertise in developing high-performance C++ and Python frameworks for surrogate modeling and Bayesian inference.

Education

Ecole Polytechnique (CMAP) and Politecnico di Milano (DAER)

Paris (FR) and Milano (IT)

Double PhD

Sept. 2023 - Oct. 2026 (Expected).

- Thesis co-supervised by Dr. Pietro Marco Congedo (Inria Saclay) and Prof. Giulio Gori (Politecnico di Milano), financed by the **MSCA project TRACES**.
- Devised a surrogate-based accelerated calibration framework for inferring the parameters of a mis-specified computer code using experimental measurements and Bayesian methods. Developed a procedure to reduce the dimensionality of the calibration problem. Devised a Domain Decomposition Algorithm to surrogate discontinuous computer codes.
- Implemented those algorithms in a self-maintained efficient C++ library (CMP++).
- Validated the calibration technique through a collaboration with the Von Karman Institute to estimate the Oxygen/Nitrogen catalytic efficiencies for Thermal Protection Materials for Hypersonic applications.
- Industrial Secondment at TerraXcube (Bolzano), developed a surrogate model of their Icing Wind Tunnel, integrated measurement uncertainty to improve robustness during predictions.
- Teacher assistant for the MSc course "Finite Element Analysis" at CNAM Paris.
- Participated in a virtual certification exercise for an Electro-thermal Ice Protection System within TRACES. Integrated industry feedback from GE Aerospace, Dassault Aviation, and Airbus to align the design with European aviation safety standards.

Politecnico di Milano

Milano (IT)

MSc Aeronautical Engineering

Sept. 2020 - May. 2023

- Final Grade 110 cum Laude, curriculum focused on aerodynamics.
- Member of the CFD team of Skyward Experimental Rocketry. Performed steady-state subsonic CFD simulations of rockets under different conditions in OpenFOAM and automated CFD calls using Python.
- Experimental thesis on electro-hydrodynamic (EAD) thrusters. Developed a theoretical model for an EAD thruster and performed a parametric experimental investigation to improve the Thrust to Power ratio and Thrust Density.
- Graduate of **Alta Scuola Politecnica**, an honors program that strengthens horizontal skills (courses on dynamic of innovation, design methods, communication and decision making). Final project: worked with a multidisciplinary team and Frandent Group SRL to design, prototype and test an electromagnetic shock absorber for agricultural machines. Performed COMSOL simulations and lab tests of the device.

Università degli studi di Padova

Padova (IT)

BSc Aerospace Engineering

Sept. 2017 - July 2020

- Final Grade 110 cum Laude.
- Member of the project OCEANS18, design of a sounding balloon for vessel identification. Development of a ML algorithm to detect ships from images and retrieve their coordinates.

Work Experience

Vertiv

Tognana (Padova, IT)

Internship - HVAC Engineer

6 months (2018-2019)

- Analysis of the thermodynamic, hydraulic, aerodynamic, acoustic, and electric performance of HVAC machines using company software and Excel.
- Development of Python code for automating calculations.

Skills

Languages: Italian (native), English (professional) and French (intermediate).

Coding & Software: Python (Numpy, Scipy, Pytorch and Scikit-learn), C++, Matlab, Git, Linux/HPC.

Numerical Engineering: CFD (OpenFOAM and SU2), FEA (Nastran/Patran), multiphysics (COMSOL).

Machine Learning: Bayesian Inference, Uncertainty Quantification, Surrogate Modeling (Gaussian Processes, Neural Networks and Polynomial Chaos Expansions), Dimensionality Reduction, Domain Decomposition and Digital Twin.

Laboratory & Design: Experimental Fluid Dynamics (EFD), High-voltage measurements, CAD (SolidWorks), Data Acquisition, manipulation and analysis.

Other: In 2024 I co-founded the project RandoCité (funded by CIUP Paris), we organized free hikes and activities around Ile-de-France.

A list of publications and a detailed description of my projects is available on my website.

References are available upon request.